REVEAL: Agentic In-Hand Manipulation for Language-Guided Target Presentation

Anonymous CVPR submission

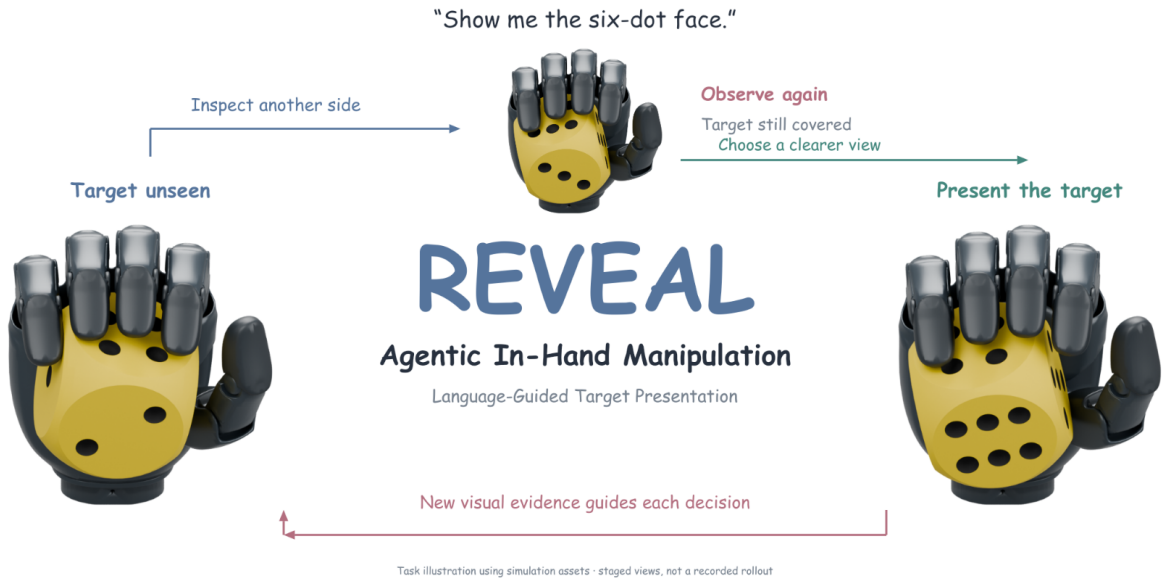


Figure 1. **REVEAL overview.** New visual evidence guides semantic decisions as the hand inspects and presents a requested object face. This staged simulation illustration is not a recorded rollout or physical-robot result.

Abstract

Language-guided in-hand manipulation requires a robot to uncover a requested object face while maintaining contact. This couples visual information gathering with motor control, yet high-level visual reasoning introduces delays during which the hand must continue acting. We present REVEAL, an asynchronous framework for online visual search and semantic target presentation. The agent uses updated images to inspect the object and bind the requested target to a shared goal bank, which interfaces with a history-conditioned motor policy. Teacher-supervised local controllers adapt joint commands during pending requests and subsequent execution. Simulation evaluations cover six object-specific systems, with additional English/Chinese instruction tests on three objects. Under changed semantic mappings, REVEAL achieves an object-averaged success rate of 65.8%, exceeding the strongest evaluated baseline by 40.4 percentage points.

1. Introduction

A command such as "present the face marked T" specifies a visual goal, not a sequence of hand motions. The robot must find the mark, associate it with a controllable orientation, and rotate the object while preserving contact. The target may be hidden by the object or the fingers, so an informative action also changes the grasp and the next observation. In-hand target presentation therefore couples visual reasoning with contact-rich control.

Vision-language agents offer a flexible way to interpret such requests, but their decisions arrive asynchronously. During inference, the hand and object continue to move. Repeating the previous command may lose the grasp or leave the target occluded; slowing motion alone may preserve contact without producing a useful correction. A practical system needs a local controller that responds to current measurements while semantic reasoning is pending.

Our key insight is to separate the rate of semantic decisions from the rate of motor control while keeping both interfaces grounded in the same goal representation. An

asynchronous visual agent inspects the object and issues a validated semantic command. A history-conditioned student maps the active goal and measured pose to joint targets. For the cube, separate local controllers adapt those targets during pending requests and ordinary execution, using privileged teacher actions for training. This lets the hand respond to its current state without waiting for another high-level call.

We instantiate this design by adapting a ViserDex-style RGB-to-pose policy [4] to a continuous semantic goal input while reusing compatible recurrent weights. The REVEAL agent emits public tool arguments that map to the shared goal bank; it does not expose hidden model embeddings. A fine-tuned Qwen3-VL baseline [2] predicts a goal vector directly, and a rule-based VLM parses a color and readability decision. These interfaces provide a common motor policy while retaining the differences between an agent with feedback and history and single-step grounding baselines.

Across 1,440 simulation trials spanning six object-specific systems and two semantic mappings, REVEAL attains object-averaged success rates of 70.4% on original mappings and 65.8% on changed mappings. These exceed the strongest evaluated baseline mean in each setting by 26.2 and 40.4 percentage points, respectively. The means describe complete systems with object-specific weights and perception, not zero-shot policy transfer. A separate 2,160-trial English/Chinese study favors REVEAL over fine-tuned grounding in all eight duck and hedgehog conditions after multiplicity correction; cube differences remain unresolved. Matched cube interventions separately test fresh visual evidence and local control.

Our contributions are threefold. First, we formulate language-guided target presentation as online visual search and semantic binding during continuous in-hand control. Second, we develop an asynchronous agent–motor interface with public semantic actions and teacher-supervised local adaptation. Third, we evaluate that interface through six-object mapping comparisons, three-object language coverage, and cube interventions that examine fresh visual evidence and control during pending requests.

Task mappings, model weights, stopping rules, and physical success criteria are fixed within each cohort. Service exclusions are applied symmetrically within comparisons, with all raw outcomes retained. We report physical reach separately from correct semantic binding and autonomous completion.

2. Related Work

RL for dexterous in-hand manipulation. Simulation-trained RL policies enable physical-hand reorientation with domain randomization [1] and general reorientation across diverse object geometries [7]. Rapid motor adaptation supports fingertip rotation across changing object properties [15]; RotateIt fuses vision, touch and proprioception through teacher–student distillation [16]. AnyRotate extends tactile control to arbitrary rotation axes and hand orientations [21]. These works establish motor capabilities on which semantic interaction can build. REVEAL studies how to discover and bind a language-specified target when its appearance is initially unavailable, rather than a new general-purpose reorientation algorithm.

Visual and goal-conditioned dexterity. ViserDex combines monocular pose estimation with recurrent control [4]; we reuse this decomposition and adapt a released Revo3 policy to semantic goals. GoalDex conditions a visuomotor policy on current and goal images, uses a VLM to select goals from language, and studies multi-object sim-to-real transfer [3]. It also models visual-processing delays. REVEAL focuses on acquiring missing evidence and binding the requested face during manipulation while local control continues. ManipTrans studies residual learning for dexterous bimanual transfer [12]; our local residual adaptation supports this semantic interaction loop.

Agentic reasoning and embodied feedback. SayCan grounds language-level skill selection in robotic affordances [11], while Inner Monologue incorporates environmental feedback into closed-loop planning [9]. Code as Policies composes perception and control APIs [14]; VoxPoser and ReKep ground language in spatial value maps and keypoint constraints [8, 10]. These works motivate separating semantic decisions from motor execution. Magma connects multimodal representations to agent actions [20], and VERA includes language-conditioned dexterous reorientation from visual plans [13]. REVEAL concentrates on an occluded target held in the hand: inspection changes available evidence, binding selects a canonical goal, and local controllers maintain manipulation during external reasoning. Our fresh-image intervention tests the role of later observations; feedback-driven planning itself is not a novelty claim.

Asynchronous robot policies. Real-time chunk execution [5], online VLA corrections [18], latency-aware RL [22], and temporal counterfactual preferences [23] address inference delays. LiMA combines sparse semantic intention with asynchronous dexterous action generation [6]. REVEAL learns compact teacher-supervised controllers around a separate motor policy and public semantic tools. Its evaluated distinction is the observation–binding–control loop; asynchronous computation and residual actions alone are not novel. The controller ablations do not isolate the benefit of explicit goal conditioning.

3. Task and Evaluation Contract

Semantic target. Q1 covers six object-specific simulation releases: a marked cube, a wooden die, a duck, a hedgehog, a pig, and a milk carton. Q2/Q3 interventions use only the marked cube. Cube and marked-object instructions identify one of six surface letters; die instructions request one through six dots. The cube letters are $\{T, E, R, L, S, H\}$. Each target has a canonical world orientation $Q(g)$ that places its marked face upward. The environment scores the full canonical quaternion, including its fixed twist, rather than accepting every orientation with the correct normal. The language prompt is therefore less specific than the scoring convention. All methods use the same scoring convention.

At step t, the agent observes RGB evidence I_t and the instruction ℓ. The motor system observes proprioception x_t and measured pose $\hat{p}_t$. The cube obtains $\hat{p}_t$ from RGB; the other releases use a noisy simulator-pose model. Clean true pose p_t and teacher actions are used for supervision and independent scoring, and future semantic responses are unavailable to online control. We denote the active goal vector by g_t, recurrent state by h_t, and the pending-request flag by m_t.

Physical time. One request may be pending at a time. The cube protocol allows six model calls per episode and 3,600 logical control steps at 30 Hz; the other objects use 20 Hz over the same 120-second logical horizon. If a call starts at k and takes L wall-clock seconds, its response becomes usable only after the call completes and at least $\lceil 30L \rceil$ control steps have elapsed. Physics continues throughout this interval. The simulator currently runs slower than real time. This protocol represents inference delay in simulated physical time, but does not demonstrate a 30 Hz hardware pipeline.

Autonomous completion. All cube systems share a measured-pose stopping gate, evaluated at 3 Hz. It requires a declared binding for the requested symbol, no pending request, a non-inspection state, and measured full-orientation error at most $\epsilon = 0.16$ radians. The gate may stop independently of an agent's native stop suggestion. The other object releases use calibrated completion correction for REVEAL and a measured-pose gate for bare baselines, with the same independently scored tolerance; their stopping conditions are documented in the supplement. At stop time τ, the independent success variable is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared face binding, q_τ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. Merely passing through a target does not count. No extra hold duration is added after stopping. A pose false stop and a semantically wrong binding are failures even if the hand retains the object. The frozen drop rule is distance of at least 0.24 metres from the nominal object grasp position.

Matched reporting. Unless stated otherwise, each arm uses 40 raw trials from two seeds, with 20 trials per seed and matching initial physical states and scored physical targets. Instructions also match within the control and feedback studies. In the remapping study, the requested letter changes with the mapping to preserve the physical target. A service error or measured model latency above 120 seconds is recorded as a service exclusion. For a panel of methods $\mathcal{C}$, the eligible set removes the union of excluded trial identifiers across $\mathcal{C}$. We report both raw and eligible outcomes, never replace an excluded trial, and never mix denominators from different panels. Long service delays remain visible without being interpreted as controller failures.

4. Method

REVEAL connects semantic decisions to motor control through a shared goal bank (Fig. 2). We detail the cube implementation below; other object releases use their own history-conditioned students, noisy pose feedback and maintenance actors, as specified in the supplement.

4.1. Reusing a visual pose-conditioned student

A frozen CNN predicts the image coordinates and depth of the object center and eight keypoints from a 120×120 RGB crop. Calibration and rigid fitting recover the measured position and XYZW orientation, together with a fit residual. Cube trials use rendered object views with live hand occlusion; deployment receives no ground-truth pose.

The inherited student expects geometric goals. We replace its goal channels with a learned projection of $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains target-free proprioceptive features and history, while $u_t \in \mathbb{R}^{21}$ gives normalized joint-target proposals. Compatible encoders, recurrent weights, actor weights, and normalization statistics are initialized from the released policy. Neither target quaternion nor relative-goal quaternion is an input to the migrated student.

We fit the new goal interface and adapt the recurrent core with frozen-teacher actions in a DAgger-style procedure [17]. Teacher state and actions are training-only. The cube deployment uses the frozen RGB pose estimator; object-specific policy and perception differences for the other releases are documented in the supplement.

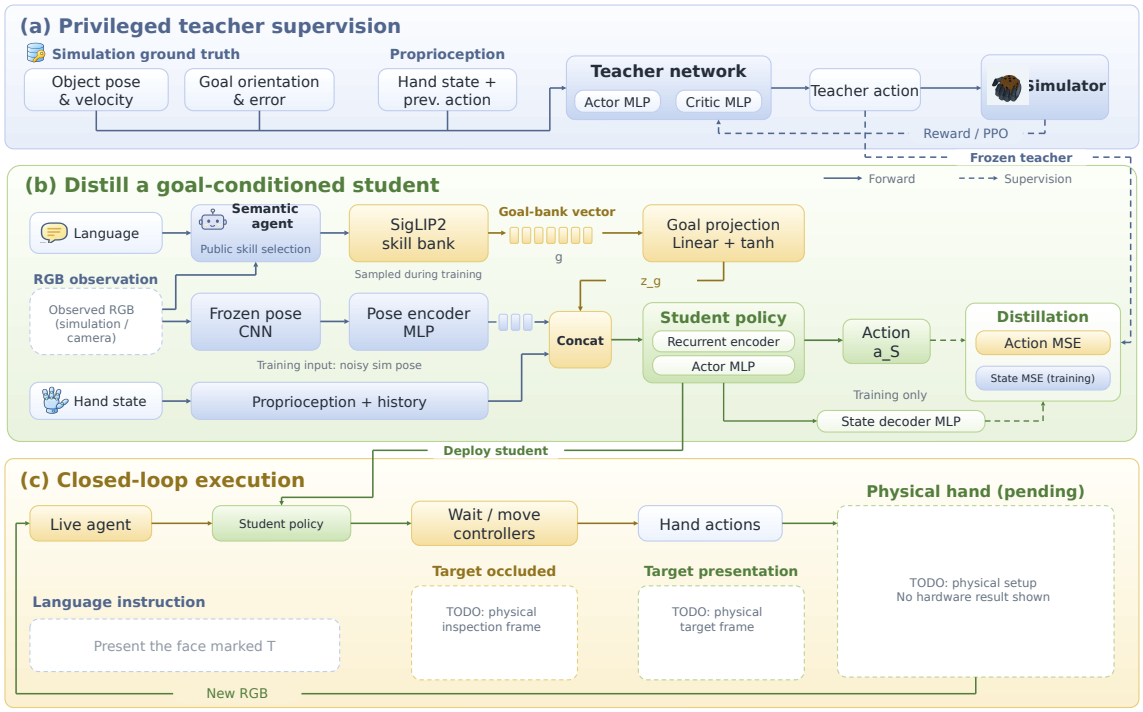


Figure 2. **REVEAL training and deployment pipeline.** Privileged supervision trains a student conditioned on a public goal-bank vector. The semantic agent selects skills, and local controllers adapt their execution. Dashed physical-hand slots are pending; they do not report hardware results.

4.2. Semantic grounding and public actions

The cube uses six normalized goal vectors $B = \{g_1, \ldots, g_6\}$ from cached SigLIP2 text features [19]; each is bound to a color skill and canonical asset orientation.

Semantic agent interface. Given the requested symbol and current image, the hosted agent emits a validated public action (inspect, reorient, verify, stop, or abstain). A color argument selects a vector from B; inspection may change the motor goal without binding the requested symbol. Agent-selected inspection/reorientation actions have bounded 10–90-step durations. Request latency is separate from action duration; updated images and public action history inform subsequent decisions. Public arguments map deterministically to the student goal; hidden agent features are not used.

4.3. Learning control while a request is pending

An outstanding query does not determine whether the current movement is still safe. We construct a deployable feature vector s_t from proprioception, measured pose, student belief, proposed action, pose-fit residual, semantic binding status, and request status. It also includes the measured pose and command at the request anchor and elapsed pending time. The active goal g_t is encoded separately. Before a binding is

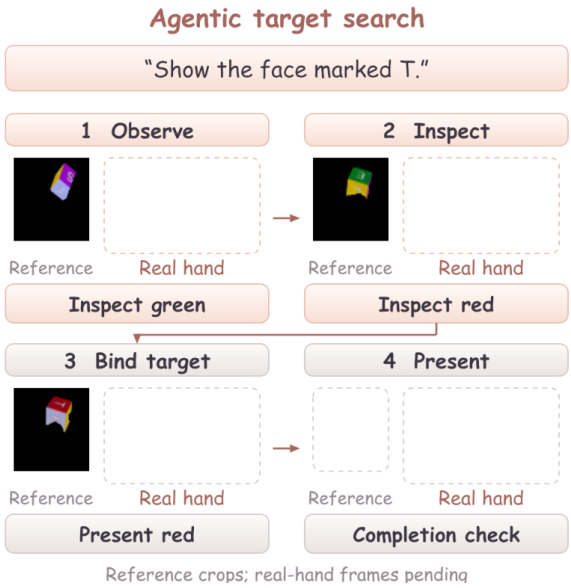


Figure 3. **Visual target search.** The agent observes, inspects candidate faces, binds the requested target, and issues a presentation command. The reference crops are simulated; real-hand frames are pending.

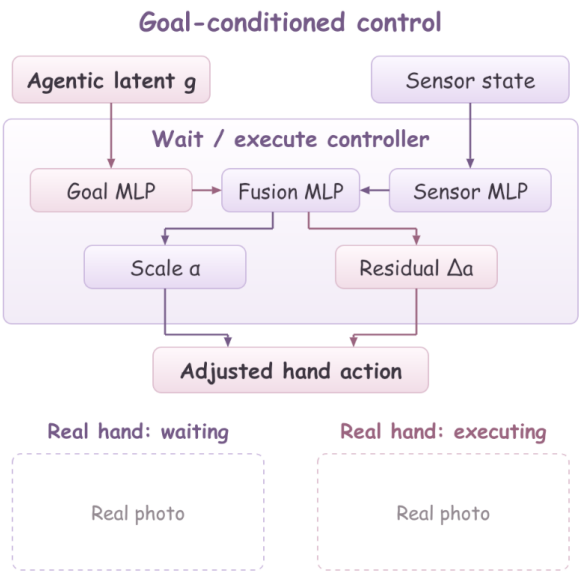


Figure 4. **Local control component.** A goal-conditioned actor combines sensor and goal features to scale the student proposal and add a bounded residual. The photo slots in the source illustration are pending and do not show real-hand results.

established it denotes the startup or inspection skill, not the unavailable future target.

A sensor encoder and a goal encoder feed a fusion layer. The waiting actor predicts a speed logit r_t and a residual $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ be normalized current joint position and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \mathrm{sigmoid}(4r_t + \mathrm{logit}(0.15)), \tag{4}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{5}$$

$$a_t = \mathrm{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{6}$$

Zero outputs reproduce fixed slowing with coefficient 0.15 while a query is pending. The environment's joint mapping and smoothing remain unchanged. The residual permits contact adjustments outside the interpolation between current joints and the student proposal.

Waiting targets are generated by the frozen teacher with the request anchor as the holding objective. This supervision is available in simulation. The deployed actor receives measurements and the active goal only. It never receives the answer being computed. The objective fits the applied action to the teacher label and penalizes residual magnitude:

$$\mathcal{L}_{\mathrm{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}}\left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2\right]. \tag{7}$$

The waiting-actor refit changes only pending-request outputs; the student, predictor, execution controller, and stopping gate remain fixed. Training details are provided in the supplement.

4.4. Learning execution after a decision

When $m_t = 0$, a separate execution actor supplies v_t in Eq. 6, using the student proposal as its base. It learns a bounded teacher-supervised correction for non-pending execution. Its active goal is explicit, but the current evaluation does not establish the benefit of this separate goal-input pathway.

4.5. Closed-loop execution

At each control step, the system updates visual pose, advances the recurrent student, chooses the waiting or execution actor, and applies the resulting joint targets. A completed semantic response is validated before it changes the active skill or declared binding. The next image reflects the motion that occurred while the previous response was pending. The common stopping gate is checked separately from semantic action generation. Invalid model output cannot supply ground-truth information or silently select the requested target.

Student adaptation and controller learning are evaluated separately because changing the student also changes the belief and action proposals received by the controllers.

5. Experiments

We organize evaluation around four questions. Q1 tests complete systems, Q2 studies local control, and Q3 tests visual feedback. Q4 specifies the pending hardware evaluation. Q2 and Q3 use only the marked cube.

Q1. How do complete systems compare across objects, mappings and instruction variants?
Q2. How should the hand act during deliberation?
Q3. Does continued visual feedback help?
Q4. Does the system work on physical objects?

5.1. Setup and reporting

Experiments use Revo3 assets in Isaac Lab 3.0/Newton with validated contact capacities. Unless specified, each condition has two evaluation seeds and 40 scheduled trials. Physical tasks are paired within each study. Cube systems share the frozen visual pose estimator, goal bank, latent student, 0.16-radian tolerance and measured-pose stopping. Other objects use separate students and noisy simulator pose; REVEAL additionally uses calibrated completion correction. Bare VLM baselines omit the added control actors and use a fixed unreadable-view schedule, while REVEAL adapts its inspection. REVEAL denotes the complete deployed system. Natural model latency and these backend differences are part of the complete-system comparison.

The latest cube release replaces only its waiting actor. Q1 and Q3 rates use common service-eligible slots across compared methods; Q2 reports all 120 outcomes per method.

Q1 and Q2 use distinct evaluation cohorts: Q1 measures complete systems across mappings, whereas Q2 isolates pending-request control on the cube. Their absolute rates are not matched estimates; controller gains are computed within Q2. The three-seed Q2 table is descriptive. Paired tests and any multiplicity corrections are specified separately for each study in the supplement.

5.2. Q1: Systems across objects and mappings

We compare original and changed semantic mappings across six object-specific releases. Historical native-image cohorts remain separate in the supplement; their REVEAL–fine-tuned differences are unresolved (cube $p = 0.267$, die $p = 0.4545$).

Aggregate system performance. Over 1,440 scheduled trials, equal-object mean success is 70.4% on original mappings and 65.8% on changed mappings. The strongest baseline means are 44.2% (fine-tuning) and 25.4% (rules), yielding gains of 26.2 and 40.4 percentage points. These post-hoc descriptive means weight six service-eligible object rates equally; they neither pool trials nor add a significance test.

Four additional objects. Duck, hedgehog, pig and milk carton contribute 960 trials across two mappings. REVEAL has the highest observed rate in each cell (Table 1); these are object-specific systems, not common-policy transfer. Three historical service failures are excluded symmetrically. Milk-carton REVEAL drops in 15/40 original and 20/40 changed-mapping trials, so higher completion does not imply stable grasping. Reset and sensitivity audits are in the supplement.

Visual-semantic intervention. Only the upper image changes; predictor input and lighting remain fixed. Identity, cyclic letter–color permutation and masking use 40 matched states per method, all with reconstructed face images. REVEAL retains 32/40 successes after permutation; fine-tuning falls from 30/40 to 16/40 (Holm $p = 0.0130$, six contrasts); rules remain at 23/40. Masking yields 0/39, 3/40 and 0/40, respectively (full supplement table). Nine of eleven fine-tuned wrong bindings after permutation select the old association, consistent with learned shortcuts but not an exclusive memorization mechanism. One permutation does not establish broad generalization or equivalence.

Dice layout transfer. A separate 240-trial study rotates only the native visual mesh, preserving collision, inertia and the prompt distinguishing printed counts from motor-skill IDs. Table 1 reports 40 matched trials per cell, with no exclusions. Permuted-layout REVEAL contrasts give exploratory paired $p = 2.98 \times 10^{-8}$ against fine-tuning and $p = 5.96 \times 10^{-8}$ against rules; its layout contrast ($p = 0.688$) does not establish equivalence. REVEAL also receives probes, history and noisy feedback, so this supports system-level transfer, not isolated model superiority.

Language and wording coverage. Three object-specific cohorts test English/Chinese standard and rewritten instructions: three evaluation seeds, 60 scheduled trials per cell, 2,160 total. Table 2 uses common service-eligible slots; weights, mappings and scoring remain fixed, without Chinese retraining. The rule baseline also localizes its task/output instructions, so this tests deployed-system coverage. Rates are conditional on service availability; the supplement retains raw outcomes.

REVEAL reaches 67.8–76.7% on duck and 73.3–76.7% on hedgehog, exceeding fine-tuning in all eight conditions (Holm-adjusted paired $p \leq 0.0258$ across 36 method contrasts). Against rules, six of eight contrasts survive correction; English-standard differences remain unresolved. On cube, REVEAL leads numerically in three conditions, while fine-tuning leads on Chinese rewriting (81.1% versus 67.9%); none of their four contrasts is significant (all adjusted $p = 1.0$). Strict pairing sensitivity preserves these decisions. No language/wording contrast survives its separate 36-test correction; this does not establish language equivalence.

5.3. Q2: Controller ablations on the cube

We compare four ways of acting while a semantic request is pending: continuing the student proposal, fixed slowing with coefficient 0.15, learned slowing, and learned slowing with a bounded joint correction. The ablation changes pending-request control while retaining the same execution controller outside pending intervals. Each method is evaluated for 120 trials, with 40 trials per seed across three seeds.

Slowing with joint correction achieves 104/120 successes (86.7%), compared with 88/120 (73.3%) for fixed slowing, a gain of 13.3 percentage points. Waiting drops decrease from 13 to five, and execution drops from eight to four. Learning the speed alone yields 82/120 successes (68.3%), below fixed slowing. These results favor combining speed adaptation with joint correction in this evaluation; failure stages describe when a drop occurs rather than identifying its cause. Seed settings and outcome breakdowns are reported in the supplement.

5.4. Q3: Visual feedback on the cube

We test two complementary interventions in separate paired cohorts (Table 4). Panel (a) compares complete feedback protocols; panel (b) isolates whether later images carry useful evidence under a common call cap. Their outcomes are not pooled.

Object	Motor pose	$n_{\text{orig}}/n_{\text{changed}}$	Original mapping			Changed mapping		
			Rule	Fine-tuned	REVEAL	Rule	Fine-tuned	REVEAL
Cube	RGB pose	40 / 40	57.5	75.0	**80.0**	57.5	40.0	**80.0**
Die	Noisy pose	40 / 40	32.5	57.5	**70.0**	2.5	0.0	**65.0**
Duck	Noisy pose	38 / 40	39.5	36.8	**65.8**	22.5	5.0	**60.0**
Hedgehog	Noisy pose	39 / 40	43.6	33.3	**71.8**	25.0	5.0	**80.0**
Pig	Noisy pose	40 / 40	52.5	40.0	**92.5**	37.5	7.5	**82.5**
Milk carton	Noisy pose	40 / 40	32.5	22.5	**42.5**	7.5	2.5	**27.5**
Mean	Object-specific	–	43.0	44.2	**70.4**	25.4	10.0	**65.8**

Table 1. **Q1: six objects and two semantic mappings.** Completion (%) uses common service-eligible trials; each cell has 40 scheduled trials. Mean weights six objects equally. Bold denotes the highest observed rate, not significance. Object-specific weights remain fixed across mappings; bare baselines omit added control modules. Cube uses reconstructed upper images and RGB pose; die rotates its visual mesh; other objects permute surface letters with noisy simulator pose. Native-image and language cohorts remain separate.

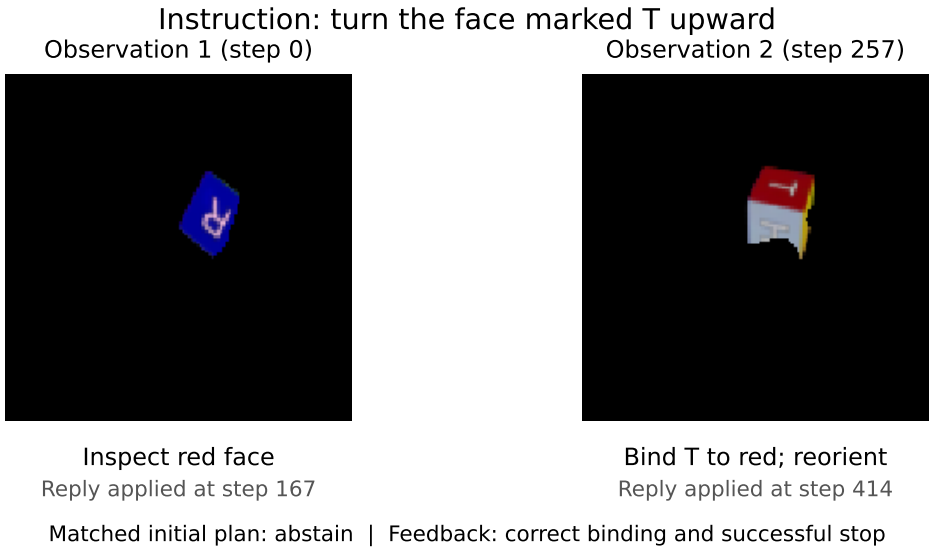


Figure 5. **Recorded inspection before binding.** Unretouched model-input frames from a recorded simulated trial. The initial plan abstains; feedback reveals T, then REVEAL binds T to red and completes at step 420 (true error 0.0812 radians). Both arms start from identical state and image. This selected example illustrates evidence acquisition, not wrong-binding correction or hardware performance.

(a) Is continued feedback useful? The initial-plan arm receives the first RGB observation and measured feedback, then commits to one to six bounded actions without later semantic observations. Its last non-abstain skill can continue until the common stopping gate or horizon. Full feedback permits up to six calls with updated observations and history. Both arms retain continuous student pose feedback, identical local controllers and the same stopping rule.

Success rises from 6/39 to 33/39 eligible trials: 27 pairs succeed only with feedback and none only with the initial plan (exact $p = 1.49 \times 10^{-8}$). Raw success is 6/40 versus 34/40; an initial-plan HTTP 503 timeout is excluded symmetrically. Of 28 eligible initial-plan abstentions, 22 succeed with feedback. Among eligible feedback trials, 36 inspect, 32 first bind after inspection, and 31 successes use multiple calls. This supports evidence acquisition before binding, not correction of an executed wrong binding. Calls total 40 versus 120, so this contrast includes differing inference budgets. Zero initial-plan drops accompany frequent abstention. Figure 5 shows a recorded example.

(b) Does the next image matter? Both arms retain the same six-call cap and trigger policy. Only upper-agent RGB changes: current images versus the exact first frame on every call. Numeric pose feedback, history, lower perception and control remain live. Forty pairs share initial states and first images, without exclusions.

Current images yield 32/40 successes versus 9/40 for repeated first frames. All 23 discordant pairs favor fresh images, a 57.5-point gain (exact $p = 2.38 \times 10^{-7}$). A post-

System	English Standard	English Rewrite	Chinese Standard	Chinese Rewrite
Cube (n)	56	59	57	53
Rule VLM	48.2	45.8	31.6	32.1
Fine-tuned VLM	71.4	72.9	75.4	**81.1**
REVEAL	**75.0**	**79.7**	**84.2**	67.9
Duck (n)	60	59	60	60
Rule VLM	50.0	25.4	33.3	31.7
Fine-tuned VLM	38.3	30.5	31.7	35.0
REVEAL	**68.3**	**67.8**	**68.3**	**76.7**
Hedgehog (n)	60	60	52	60
Rule VLM	46.7	38.3	36.5	36.7
Fine-tuned VLM	38.3	31.7	34.6	25.0
REVEAL	**73.3**	**76.7**	**75.0**	**76.7**

Table 2. **Q1: language and wording coverage.** Completion (%) on common service-eligible trials. Methods share the listed n of 60 scheduled slots. Bold denotes the highest observed rate, not significance. Raw counts and paired analyses are in the supplement.

Waiting policy	Success	%	Wait drops	Exec. drops	Other
Continue student	73/120	60.8	24	14	9
Fixed slowing	88/120	73.3	13	8	11
Learned slowing	82/120	68.3	7	14	17
Slowing + correction	**104/120**	**86.7**	**5**	**4**	**7**

Table 3. **Q2: controller ablation on the cube.** Each method has 120 trials across three seeds. Wait and execution drops denote failure stages; Other contains the remaining failures. Outcome counts sum to 120 in each row.

Condition	Success	%	Drops	Calls
(a) Continued feedback: one vs. up to six calls				
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120
(b) Fresh visual evidence: same six-call cap				
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98

Table 4. **Q3: feedback and fresh evidence on the cube.** Each panel has 40 scheduled trials per arm. (a) excludes one service-affected pair; its full-feedback arm reuses the initial 40-trial Q2 subset. (b) has no exclusions and changes only upper-agent RGB. Drops and calls are raw totals. Cohorts are separate; realized calls are not matched compute.

hoc audit finds correct terminal bindings in 34/40 versus 9/40 trials. Abstentions fall from 24 to two, while drops change from four to three. This pattern supports acquiring target evidence rather than a gain explained solely by fewer drops. Transport hashes verify delivered images. Actual calls, tokens and trajectories differ, so the common cap does not equalize realized compute.

5.5. Q4: Multi-object hardware evaluation

Hardware protocol. **[TODO: Complete the multi-object physical-hand evaluation.]** No hardware outcomes are available yet. The planned comparison uses REVEAL and two baselines per object, reporting completion, drops, false stops and inference cost. Calibration, resets, independent scoring and synchronized-video requirements are specified in the supplement.

Supporting cube perception study. On 6,144 rendered observations, RGB-D reaches 64.73% pose recall versus 37.53% for RGB at 10 mm/10°. Ideal masks and virtual depth limit this estimator comparison; downstream gains and physical-camera accuracy remain untested. The supplement details its protocol and occlusion analysis.

6. Discussion and Limitations

What the evidence establishes. Fresh upper-level images improve cube completion with other feedback retained. Q1 does not isolate the semantic model from inspection schedules, control actors or stopping gates. Two languages and two templates do not establish broad multilingual generalization or equivalence.

What remains unresolved. The three-seed controller evaluation favors learned slowing with correction. Aggregate-only records for two seeds prevent seed-level variability estimates and paired significance testing. Explicit controller goals provide no demonstrated benefit; comparison with a generic time-aware residual controller remains open. Matched maximum calls do not imply equal realized compute, and Q2/Q3 mechanisms are tested only on the cube.

Deployment scope. Each object has separate weights. Cube pose comes from rendered RGB; other objects use noisy simulator pose. These experiments establish neither real-camera robustness nor zero-shot policy transfer. Only one changed mapping is tested per object. Full-quaternion scoring additionally fixes a twist not specified by the language request. Hosted-model revisions limit replication, and service-conditional rates do not measure end-to-end availability. Hardware evaluation remains pending.

7. Conclusion

REVEAL connects visual search and semantic binding to continuous in-hand control. Across six object-specific systems, mean success reaches 70.4% on original mappings and 65.8% on changed mappings. Cube interventions support fresh observations and waiting-time joint correction. Physical-hand deployment remains to be validated.

LLM Usage Statement

OpenAI Codex assisted with research discussion, implementation, experiment orchestration, literature retrieval, statistical analysis, figure-layout code, and drafting and editing throughout the manuscript. A hosted vision-language model and Qwen3-VL are evaluated system components. Numerical claims are grounded in recorded experiments; unfinished work is identified explicitly. The authors are responsible for verifying the text, references, code, figures, and results before submission.

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